PN-03

Object Identification via UAV Camera

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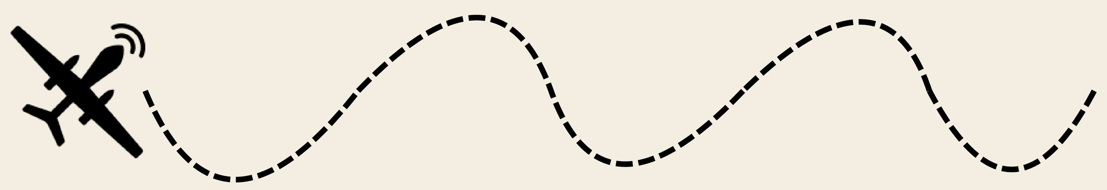
Tern Robotics

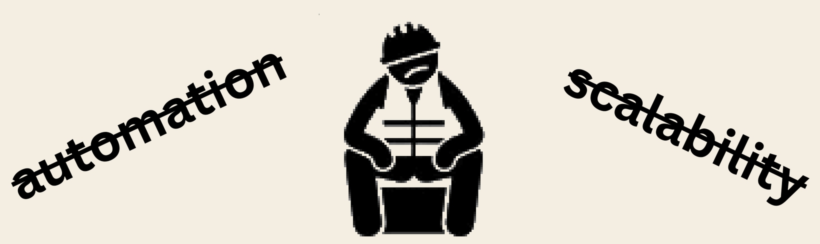
OUR CLIENT

Tern Robotics builds long-range UAVs for remote, connectivity-limited environments



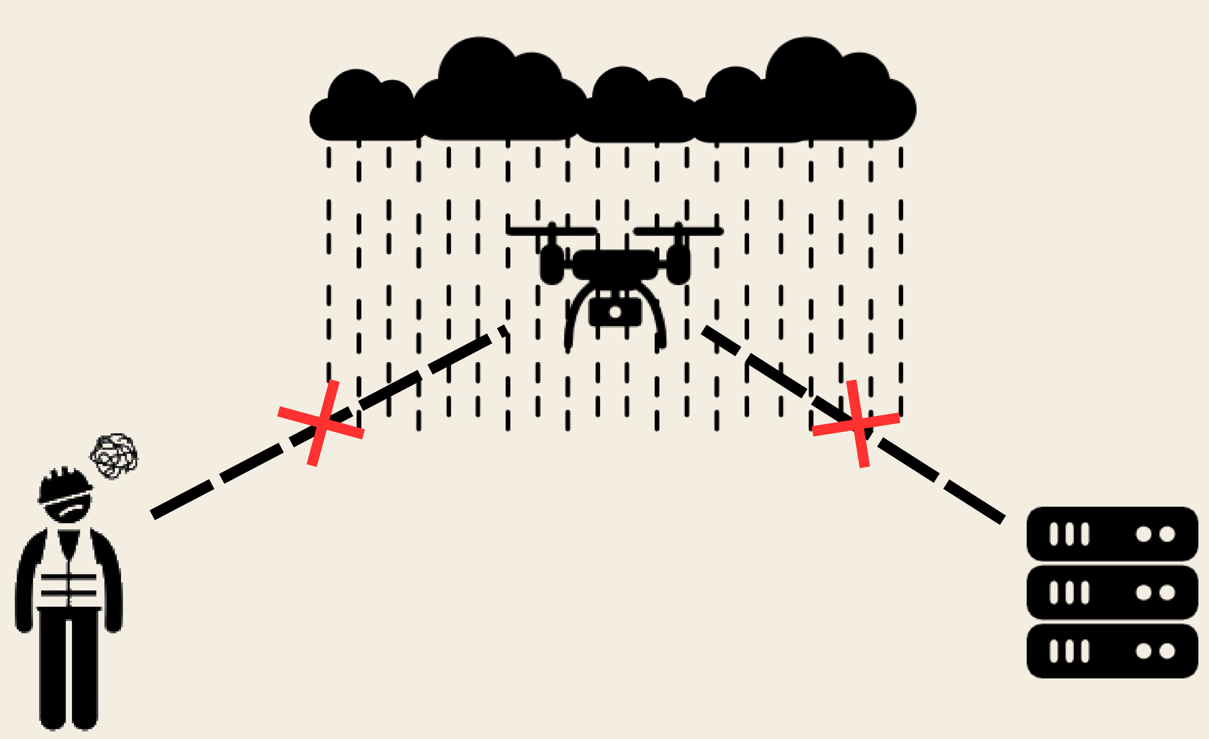
PROBLEM

Despite having cameras, operators still need to interpret footage. This slows response times and narrows mission capability.



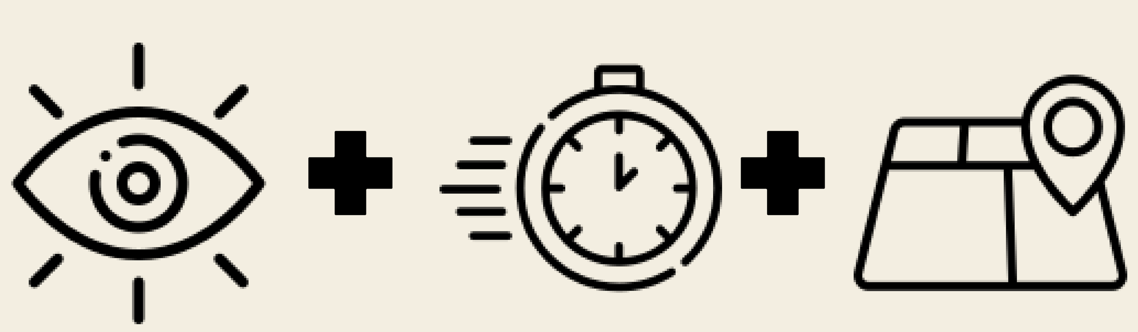
CHALLENGES

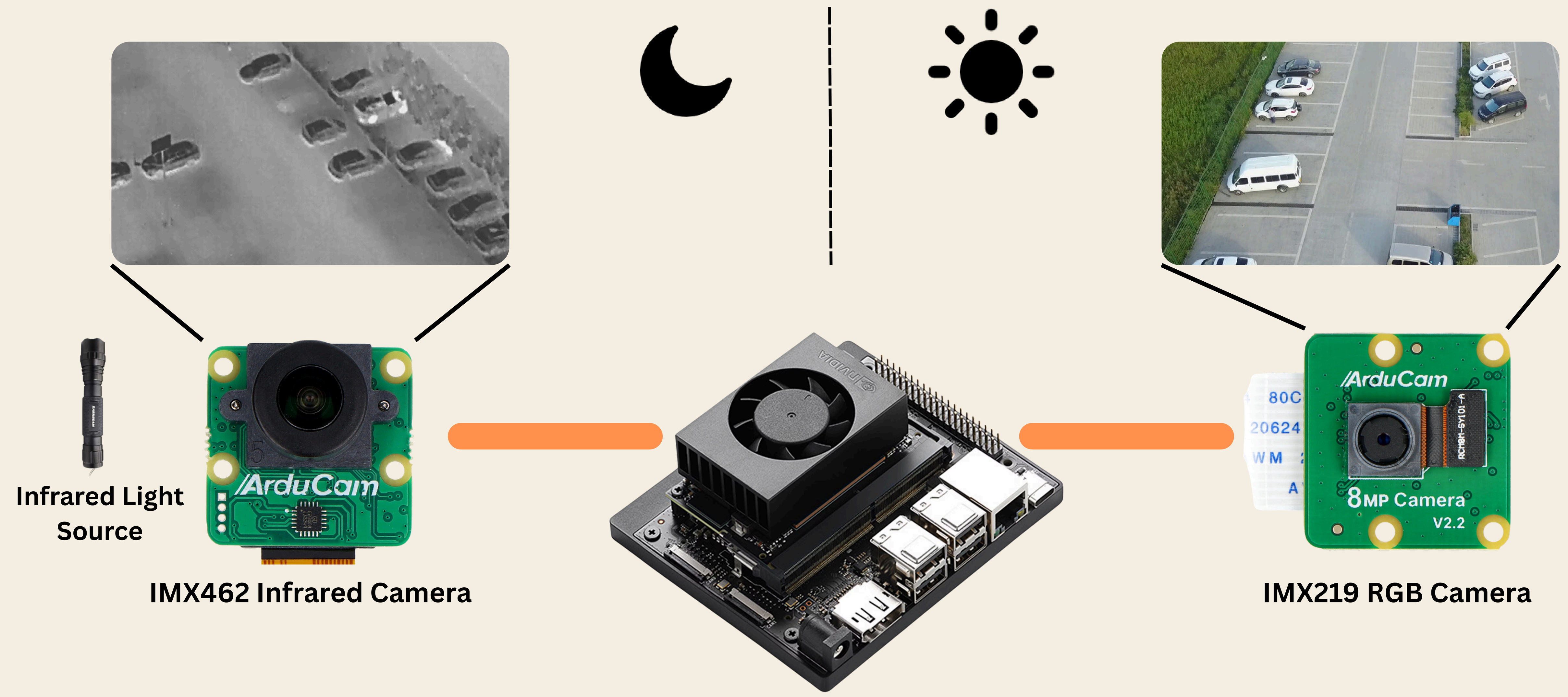
- UAVs move and change altitudes
- Harsh weather and low-light conditions
- Unreliable communication
- Cloud processing results in high-latency



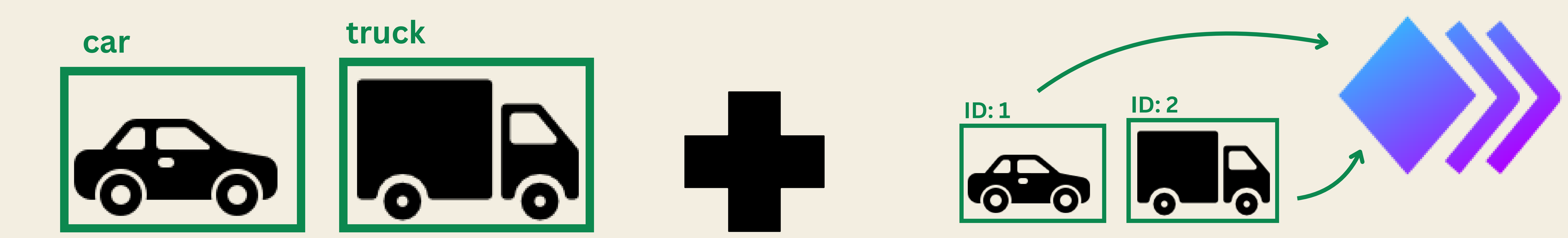
WHY THIS MATTERS

- Supports time-sensitive missions
- Improves real-time situational awareness
- Reduces operator workload
- Extends operation across diverse environments





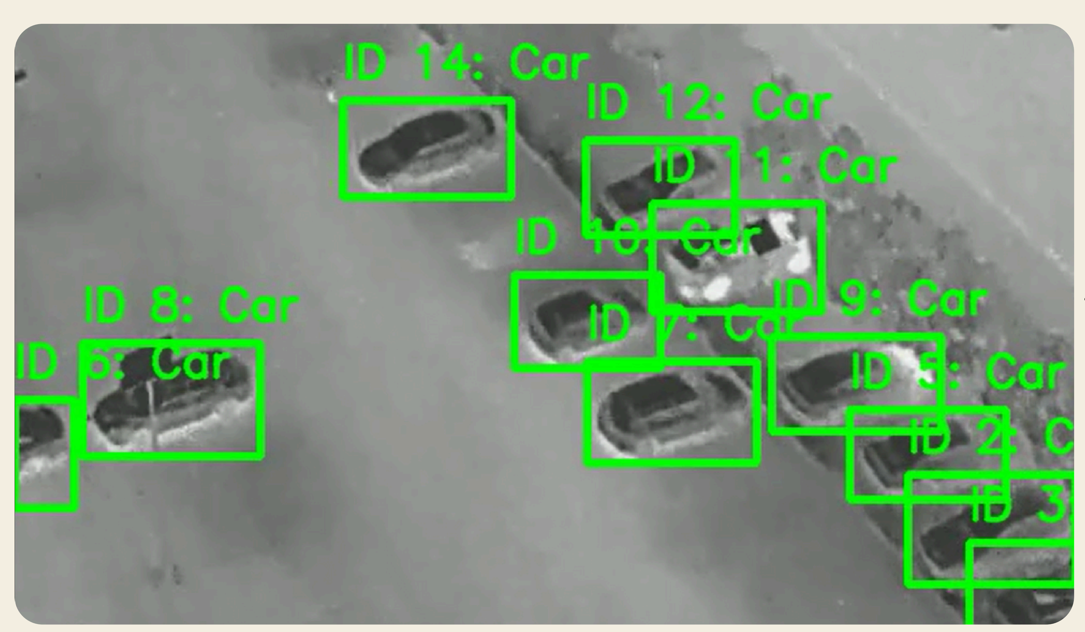
live inference

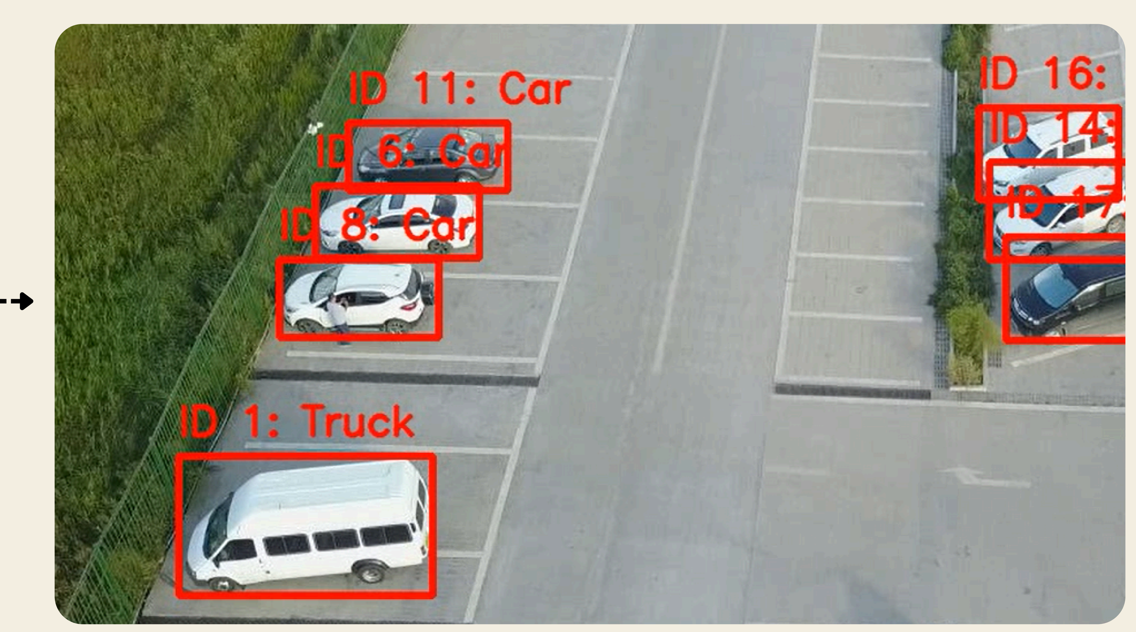


RF-DETR Nano trained on large datasets detects different objects

ByteTrack assigns IDs to objects to track them across multiple frames

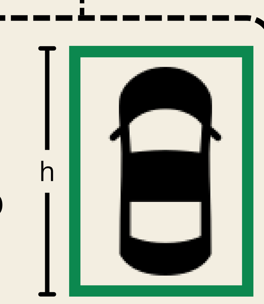
output



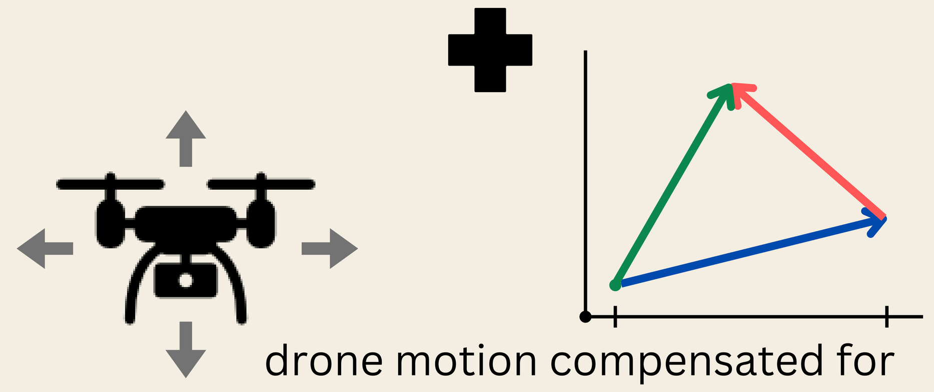


speed estimations

object and bounding box dimensions gives **meter : pixel** ratio

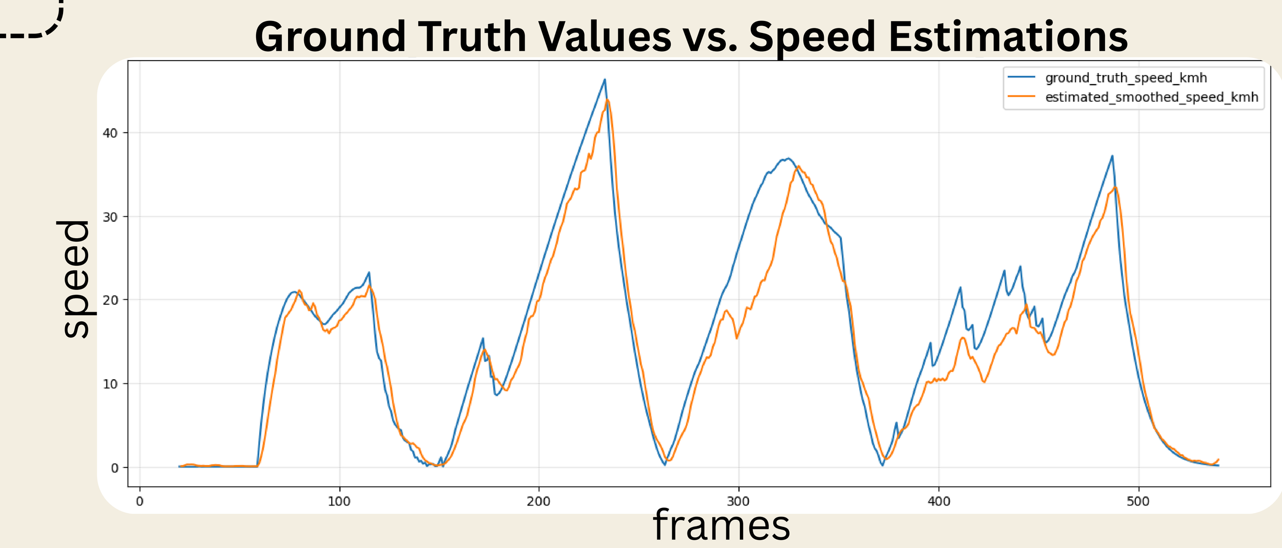


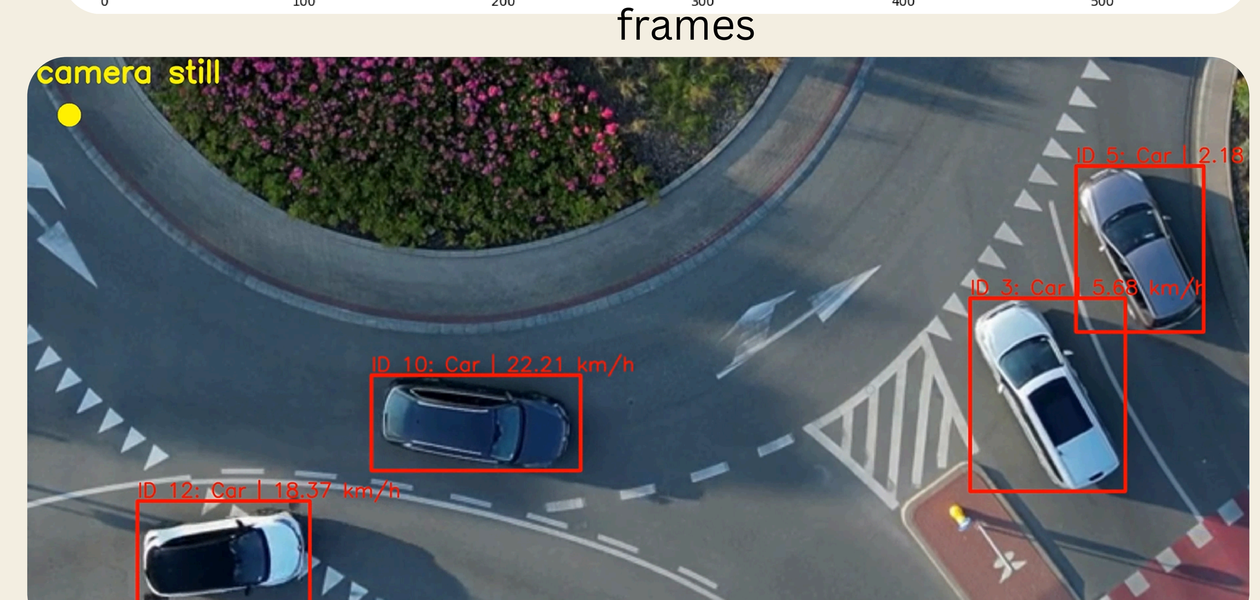
$$\Delta t = \frac{1}{FPS}, \frac{\Delta pixels \times (meter : pixel)}{\Delta t} = speed$$



drone motion compensated for using **vector additions**

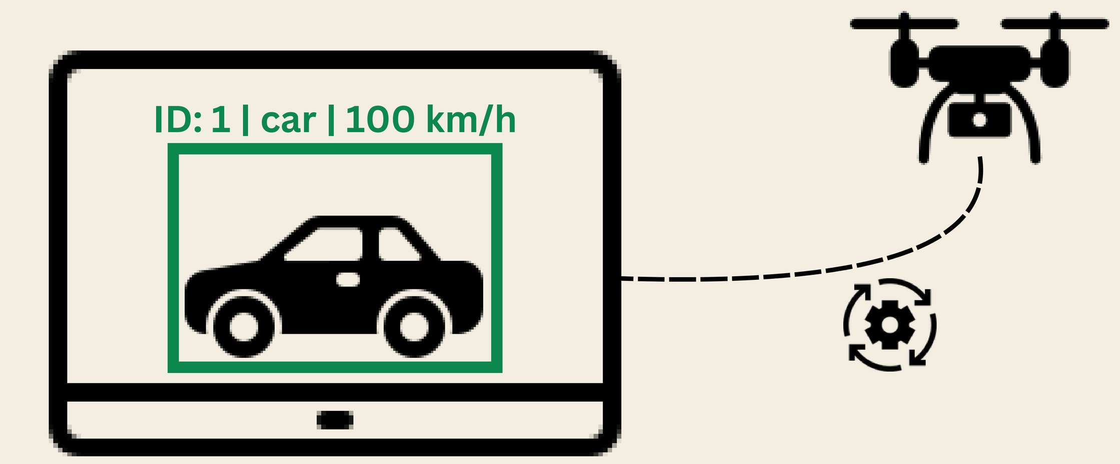
Ground Truth Values vs. Speed Estimations





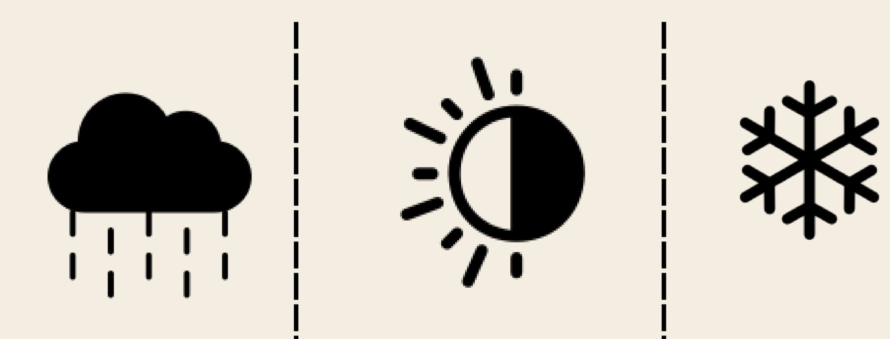
SOLUTION & CAPABILITIES

Automatic



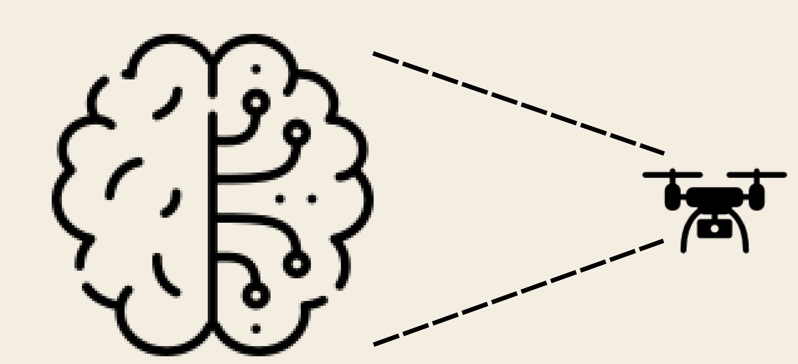
Detects, tracks, and estimates speeds of objects in real time without manual labour

Versatile



Uses RGB and infrared sensing to overcome different environments and lighting conditions

Onboard



Runs directly on the UAV, enabling fast responses during remote missions

Scalable

Built with open-source and reusable training and experimentation pipelines. Together, these act as a foundation that Tern Robotics can continue building upon.